Design of Platooning Controllers That Achieve Collision Avoidance by External Positivity

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***Abstract*— This paper solves the problem of designing local adaptive cruise controllers that guarantee collision avoidance of the vehicles in a platoon. Whereas previous studies have solved this problem for specific vehicle models, the present paper shows a general solution for a wider class of linear vehicle models. In particular, the problem of finding a controller that renders a vehicle externally positive to guarantee string stability and collision avoidance is studied in detail. The result is an algorithm that finds a set of closed-loop eigenvalues that correspond to an externally positive feedback loop. The developed methods are applied to an example and the effectiveness is verified in a laboratory experiment with multiple robots driving in a platoon.**

***Index Terms*— Adaptive cruise control, vehicle platooning, externally positive systems, collision avoidance, string stability.**

I. Introduction

A. Vehicle Platooning

VEHICLE platooning describes the problem of arranging a set of vehicles in a line so that the inter-vehicle gaps match a predefined (possibly velocity dependent) spacing. Modern vehicles come with several driver assistance systems. One of them is called adaptive cruise control (ACC) which should solve the platooning problem from a local perspective. For this feedback loop, collision avoidance is a crucial requirement that has to be ensured by each individual vehicle in the platoon.

In [1], it has been proven that the controlled vehicles have to be *externally positive* to guarantee collision avoidance. This property ensures that the transients of each controlled vehicle follow a stepwise reference monotonically, i. e. without overshooting. Multiple studies with commercially available ACC systems showed that the controlled vehicles are not externally positive [2] and would end up in a collision due to their hazardous behaviour if the driver had not intervened [3], [4]. Thus, the present paper concerns the design of ACC systems that achieve collision avoidance by external positivity without manual intervention of the vehicle's driver. The predecessor following scheme, which is typical for ACC platoons, is applied to achieve collision avoidance on a fundamental level that does not require digital communication. The proposed methods can be extended to cooperative ACC (CACC) in order to improve performance and disturbance attenuation [5].

Manuscript received 3 March 2021; revised 5 August 2021 and 22 October 2021; accepted 7 December 2021. Date of publication 22 December 2021; date of current version 12 September 2022. The Associate Editor for this article was R. Malekian. *(Corresponding author: Alexander Schwab.)*

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Digital Object Identifier 10.1109/TITS.2021.3134803

B. Literature Overview

The main scope of this paper includes, on the one hand, the design of platooning controllers that satisfy local design goals, which in turn guarantee a desired platoon behaviour and, on the other hand, the general problem of designing externally positive feedback loops. Some relevant studies concerning these two topics are categorised in the following.

Control of Vehicle Platoons: Vehicle platooning has been shown to potentially increase efficiency and safety in daily traffic [6]. One of the earliest works on this topic considered centralised optimal control techniques [7]. However, due to the decentralised character of traffic problems, decentralised methods to satisfy safety requirements are necessary for practical applications. Thus, in [8], the concept of *string stability* has been introduced which was applied in various theoretical and practical studies [9]–[13]. All these publications use a specific vehicle model with integrating dynamics and apply weak string stability to ensure safety requirements by shaping the closed-loop transfer function so that its magnitude is bounded. However, as shown in [1], the stricter property of external positivity has to be imposed on the controlled vehicles to guarantee collision avoidance. Thus, the present paper considers external positivity as a safety requirement for vehicle platoons which implies strong string stability because an externally positive vehicle is also $\mathcal{L}_\infty$ string stable [14].

Design of Externally Positive Control Loops: A comprehensive monograph on positive systems is [15]. The design of externally positive feedback loops is a rather difficult problem since external positivity is characterised by a nonnegative impulse response which can be hardly determined analytically. Thus, the general design problem is unsolved [16]. However, a constructive sufficient condition for external positivity based on a factorisation of the transfer function was published in [17]–[19] and will be applied in the present paper.

C. Contribution and Structure

The present paper solves the ACC design problem for a general class of linear vehicle models of arbitrary order and contributes to the research on vehicle platooning and on the design of externally positive control loops as follows: The design of adaptive cruise controllers is considered for a wider class of linear vehicle models than in the recent literature and generalises our previous work [20], where an ACC design procedure was developed for a very specific first-order vehicle model. Furthermore, a novel design procedure to achieve externally positive controlled vehicles based on a state feedback is proposed. To this end, a procedure to find a set

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of closed-loop eigenvalues that correspond to an externally positive feedback loop, which can be placed by a state feedback via Ackermann's formula, is given by Algorithm 1. The main result of this paper is given by Algorithm 2, which summarises the overall design procedure.

The paper is organised as follows. Section II describes the platooning problem, introduces the models used in the paper and formalises the corresponding controller design problem. In Section III, the design problem is solved by a design procedure that renders the controlled vehicle externally positive. An application example and an experimental evaluation are given in Sections IV and V, respectively, and the paper closes with a summary in Section VI.

D. Notation

Matrices and vectors are typeset in boldface (e.g. $\boldsymbol{A}$ or $\boldsymbol{x}(t)$). The matrices $\boldsymbol{I}$ and $\boldsymbol{O}$ denote the identity matrix and the zero matrix, respectively. Their dimensions are assumed to be appropriate to the context. Functions of time t are denoted by lower-case letters (e.g. $g(t)$) and the corresponding function in the Laplace domain by the same upper-case letters (e.g. $G(s)$). The Laplace transform is symbolized by $\circ\!\!-\!\!\bullet$ as in the example $g(t) \circ\!\!-\!\!\bullet G(s)$. Symbols that characterise the closed-loop or the open-loop system have a bar or a subscript zero, respectively (e.g. $\bar{\boldsymbol{A}}$ or $\boldsymbol{A}_0$).

II. Control of Vehicle Platoons

A. Requirements on the Platoon Behaviour

This paper considers a set of N vehicles driving one behind the other. The vehicles are numbered in ascending order $1, 2, \ldots, N$ beginning with the leader. Each follower $i > 1$ is equipped with sensors to measure the inter-vehicle distance

$$d_i(t) := s_{i-1}(t) - s_i(t) - d_0, \quad i = 2, 3, \ldots, N \tag{1}$$

to its respective predecessor with $s_i(t)$ denoting the position of vehicle i and d_0 a minimum distance which includes the vehicle length and a least required separation between two consecutive vehicles (Fig. 1). The leader vehicle $i = 1$ is assumed to follow its constant reference velocity v_0 with an appropriate velocity controller. The platoon should satisfy the following requirements [1]:

(R1) Asymptotic synchronisation: Stationarily, all followers should travel with the same velocity as the leader, i.e.

$$\lim_{t\to\infty} |v_i(t) - v_1(t)| = 0, \quad i = 2, 3, \ldots, N \tag{2}$$

should hold with $v_i(t)$ denoting the velocity of vehicle i.

(R2) Asymptotic time headway spacing: The desired inter-vehicle distance should increase with the velocity and the distances should satisfy the requirement

$$\lim_{t\to\infty} |d_i(t) - \beta v_i(t)| = 0, \quad i = 2, 3, \ldots, N \tag{3}$$

with β denoting the time-headway coefficient.

(R3) Continuous progression: There should be no situation in which a vehicle moves backwards, i.e.

$$v_i(t) \geq 0, \quad t \geq 0, \quad i = 1, 2, \ldots, N. \tag{4}$$

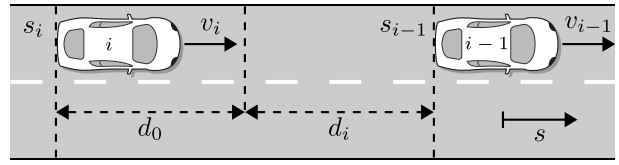

Fig. 1. Two consecutive vehicles.

(R4) Collision avoidance: All inter-vehicle distances should be nonnegative

$$d_i(t) \geq 0, \quad t \geq 0, \quad i = 2, 3, \ldots, N. \tag{5}$$

These requirements should be satisfied with local vehicle controllers. The way to find these controllers is as follows. After introducing the considered class of vehicle models and the control structure, the requirements **(R1)–(R4)** on the platoon will be broken down to design requirements on the individual vehicles. Section III presents, as the main result of this paper, a design method for the local vehicle controllers.

B. Vehicle Model

The velocity $v_i(t)$ of vehicle i is determined by the strictly proper n-th order state-space model

$$\Sigma_i : \begin{cases} \dot{\boldsymbol{x}}_i(t) = \boldsymbol{A}\,\boldsymbol{x}_i(t) + \boldsymbol{b}\,u_i(t) \\ v_i(t) = \boldsymbol{c}^{\mathrm{T}}\boldsymbol{x}_i(t), \quad \boldsymbol{x}_i(0) = \boldsymbol{x}_{i0}, \end{cases} \tag{6}$$

which will be referred to as the *vehicle model*. Its state $\boldsymbol{x}_i(t) \in \mathbb{R}^n$ may represent (but is not limited to) physical quantities of the power train, e.g. acceleration, jerk or torque. The scalar input $u_i(t)$ is, for example, the commanded acceleration of the power train. For $\boldsymbol{x}_{i0} = \boldsymbol{0}$, the input-output behaviour of Σ_i is equivalently described by

$$\Sigma_i : \; V_i(s) = G(s)\,U_i(s) \tag{7}$$

with the Laplace transforms $V_i(s) \bullet\!\!-\!\!\circ\, v_i(t)$, $U_i(s) \bullet\!\!-\!\!\circ\, u_i(t)$ and the transfer function

$$G(s) = \boldsymbol{c}^{\mathrm{T}}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{b}. \tag{8}$$

Note that the model parameters $(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{c}^{\mathrm{T}})$ do not have to be identical for each vehicle. For the sake of readability, an individual index i is omitted throughout this paper. Furthermore, the following two assumptions are imposed on Σ_i:

Assumption 1: The state-space model (6) is a minimum realisation of $G(s)$, i.e. Σ_i is completely controllable and observable. Consequently, the controllability matrix is regular:

$$\boldsymbol{S} = \left(\boldsymbol{b}\ \boldsymbol{A}\boldsymbol{b}\ \ldots\ \boldsymbol{A}^{n-1}\boldsymbol{b}\right), \quad \operatorname{rank} \boldsymbol{S} = n \tag{9}$$

Assumption 2: Σ_i has no transmission zeros, i.e. there does not exist any frequency μ that satisfies $G(\mu) = 0$.

Due to Assumption 2, the transfer function is of the form

$$G(s) = \frac{1}{s^p} \cdot \frac{b_0}{\sum_{j=0}^{n-p} a_j s^j}, \quad 0 \leq p \leq n, \tag{10}$$

which represents a series interconnection of p integrators and an $(n-p)$-th order lag system. Apart from these assumptions, the coefficients b_0 and a_j are arbitrary. The models

$$G(s) = \frac{1}{s(\tau s + 1)} \quad \text{and} \quad G(s) = \frac{1}{ms + c} \tag{11}$$

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are often encountered in the literature, e. g. in [10] and [20], which are much more specific than the vehicle model used here. In the first model, the control input $u_i(t)$ influences the acceleration through a lag with the time constant τ and in the second model, the control input represents an accelerating force that works on the lossy vehicle with the mass m and the friction constant c. Both models comply with Assumption 2.

Note that there are studies that consider nonlinear vehicle dynamics, e. g. [21] where a velocity dependent friction $\zeta(|v_i(t)|)$ is applied to model the longitudinal dynamics:

$$m\,\dot{v}_i(t) = \zeta(|v_i(t)|)\,v_i(t) + u_i(t). \tag{12}$$

To apply the proposed methods in the present paper to achieve external positivity, it is possible to linearise the nonlinear dynamics (12) to achieve a linear model of the form (11).

C. Model of the Controlled Vehicle

The coupling of the vehicles is described by the dynamics

$$\dot{d}_i(t) = v_{i-1}(t) - v_i(t) \tag{13}$$

of the distance and, furthermore, the controller state

$$\dot{z}_i(t) = \beta v_i(t) - d_i(t) \tag{14}$$

is introduced in order to achieve the required time headway spacing II-A. The right-hand side of (14) represents the time headway error, which should be equal to zero in the steady state. The combination of the vehicle model (6), the distance dynamics (13) and the regulator state (14) yields the open-loop state equation

$$\begin{pmatrix}\dot{\boldsymbol{x}}_i(t)\\ \dot{d}_i(t)\\ \dot{z}_i(t)\end{pmatrix} = \underbrace{\begin{pmatrix}\boldsymbol{A} & \boldsymbol{0} & \boldsymbol{0}\\ -\boldsymbol{c}^{\mathrm{T}} & 0 & 0\\ \beta\boldsymbol{c}^{\mathrm{T}} & -1 & 0\end{pmatrix}}_{\boldsymbol{A}_0}\begin{pmatrix}\boldsymbol{x}_i(t)\\ d_i(t)\\ z_i(t)\end{pmatrix} + \underbrace{\begin{pmatrix}\boldsymbol{b}\\ 0\\ 0\end{pmatrix}}_{\boldsymbol{b}_0} u_i(t) + \underbrace{\begin{pmatrix}\boldsymbol{0}\\ 1\\ 0\end{pmatrix}}_{\boldsymbol{e}} v_{i-1}(t). \tag{15}$$

The control law

$$u_i(t) = -\boldsymbol{k}^{\mathrm{T}}\begin{pmatrix}\boldsymbol{x}_i(t)\\ d_i(t)\\ z_i(t)\end{pmatrix} = -\left(\boldsymbol{k}_{\mathrm{x}}^{\mathrm{T}}\ k_{\mathrm{d}}\ k_{\mathrm{z}}\right)\begin{pmatrix}\boldsymbol{x}_i(t)\\ d_i(t)\\ z_i(t)\end{pmatrix} \tag{16}$$

is a feedback of the state $\boldsymbol{x}_i(t)$ of the vehicle (6), the inter-vehicle distance $d_i(t)$ and the controller state $z_i(t)$ and leads to the model

$$\bar{\Sigma}_i:\ \begin{cases}\begin{pmatrix}\dot{\boldsymbol{x}}_i(t)\\ \dot{d}_i(t)\\ \dot{z}_i(t)\end{pmatrix} = \bar{\boldsymbol{A}}\begin{pmatrix}\boldsymbol{x}_i(t)\\ d_i(t)\\ z_i(t)\end{pmatrix} + \boldsymbol{e}\,v_{i-1}(t)\\ v_i(t) = \underbrace{\left(\boldsymbol{c}^{\mathrm{T}}\ 0\ 0\right)}_{\bar{\boldsymbol{c}}^{\mathrm{T}}}\begin{pmatrix}\boldsymbol{x}_i(t)\\ d_i(t)\\ z_i(t)\end{pmatrix}\end{cases} \tag{17}$$

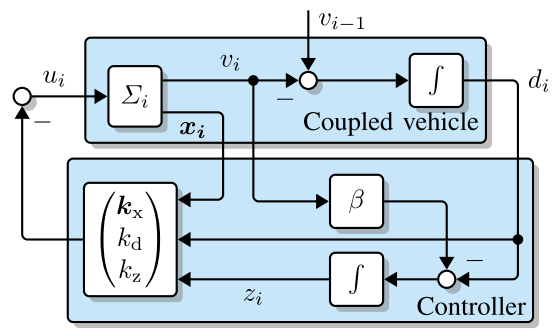

Fig. 2. Structure of the feedback loop.

of the controlled vehicle with $\bar{\boldsymbol{A}} = \boldsymbol{A}_0 - \boldsymbol{b}_0\boldsymbol{k}^{\mathrm{T}}$ and $\dim\bar{\boldsymbol{A}} = n+2$. In the frequency domain, the controlled vehicle is described by

$$\bar{\Sigma}_i:\ V_i(s) = \bar{G}(s)\,V_{i-1}(s) \tag{18}$$

with $\bar{G}(s) = \bar{\boldsymbol{c}}^{\mathrm{T}}(s\boldsymbol{I} - \bar{\boldsymbol{A}})^{-1}\boldsymbol{e}$.

Implementation of the Controller: The local feedback loop is illustrated in Fig. 2. The controller works with local quantities and without digital communication, because the distance $d_i(t)$ is measured with appropriate sensors. Thus, in order to implement the proposed control structure, the state $\boldsymbol{x}_i(t)$ has to be known. For that, the state either has to be measured directly or it has to be reconstructed with an observer or estimator. In the latter case, there are well studied methods to observe the state with a Luenberger observer (cf. [22]) or a Kalman filter, which has been applied to estimate the state of a vehicle in [23].

The measurement of the distance $d_i(t)$ is here assumed to be instantaneous. However, if the controller receives the distance information with the delay t_Δ, it has to be made sure that the time-headway coefficient β exceeds the delay

$$\beta > t_\Delta \tag{19}$$

to apply the method proposed in this paper. Due to the hierarchical coupling given by (13), the stability of the overall platoon is not jeopardised. A thorough discussion on this topic is given in [24] (Section 5.4).

D. Design Problem

The requirements **(R1)–(R4)** on the vehicle platoon are satisfied if each individual controlled vehicle $\bar{\Sigma}_i$ satisfies the following objectives:

- **(D1) Asymptotic stability:** The controlled vehicles $\bar{\Sigma}_i$ should be asymptotically stable.
- **(D2) External positivity:** The closed-loop impulse response of each vehicle should be nonnegative:

$$\bar{g}(t) = \bar{\boldsymbol{c}}^{\mathrm{T}}\mathrm{e}^{\bar{\boldsymbol{A}}t}\boldsymbol{e} \overset{!}{\geq} 0, \quad t \geq 0 \tag{20}$$

Consequently, the design problem, which will be solved in the present paper, is summarised as follows.

Problem 1: Find a gain $\boldsymbol{k}$ for the state feedback (16), so that the controlled vehicle (17) is asymptotically stable and externally positive.

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Any platoon consisting of vehicles with this controller satisfies the requirements **(R1)**–**(R4)** because for asymptotically stable controlled vehicles $\bar{\Sigma}_i$, the transients will settle:

$$\lim_{t\to\infty} \dot{d}_i(t) = \lim_{t\to\infty} v_i(t) - v_{i-1}(t) = 0 \tag{21}$$

$$\lim_{t\to\infty} \dot{z}_i(t) = \lim_{t\to\infty} \beta v_i(t) - d_i(t) = 0. \tag{22}$$

Hence, the controlled vehicles have the DC gain

$$\bar{G}(0) = 1 \tag{23}$$

and the requirements **(R1)** and **(R2)** are satisfied. In [1], it has been proven that the controlled vehicles $\bar{\Sigma}_i$ furthermore have to be *externally positive* in order to meet the requirements **(R3)** and **(R4)**.

Interpretation: An externally positive vehicle will respond with a nonnegative velocity and distance if the predecessor's velocity is nonnegative. Consequently, a platoon that is composed of externally positive vehicles will operate safely without intervention of the driver in any complex traffic condition if the velocity $v_1(t)$ of the leader $i = 1$ is nonnegative. In contrast, commercially available ACC systems that are designed to be $\mathcal{L}_2$-string stable do not have this property and, hence, often require intervention of the driver in stop-and-go traffic conditions [25].

Note that the choice of the controller structure in Fig. 2 is based on the following observation. It has been proven that it is not possible to render a feedback system externally positive using the standard control loop if the open-loop model has two poles equal to zero [16]. Since the open-loop model (15) has two eigenvalues equal to zero (as can be verified from the structure of the matrix $\boldsymbol{A}_0$), it is, therefore, not possible to achieve external positive closed-loop dynamics with a feedback of the velocity $v_i(t)$ and the distance $d_i(t)$ only. Thus, a state feedback approach is applied in the present paper, which offers more degrees of freedom than the standard control loop and renders the design problem solvable. The following section will provide an explicit solution to Problem 1 by pole placement.

III. Controller Design

A. Way of Solution

The design objectives **(D1)** and **(D2)** will be achieved by the factorisation approach presented in [17]–[19]. The idea is to find a closed-loop transfer function

$$\bar{G}(s) = \kappa \prod_{i=1}^{\tilde{n}} \bar{G}_i(s), \quad \tilde{n} \le n+2, \quad \kappa > 0, \tag{24}$$

which represents a series connection of $\tilde{n}$ subsystems $\bar{G}_i(s)$. If there exists a factorisation (24) with

$$\bar{G}_i(s) \bullet\!\!-\!\!\circ\, \bar{g}_i(t) \ge 0, \quad t \ge 0, \quad i = 1, 2, \ldots, \tilde{n}, \tag{25}$$

the transfer function $\bar{G}(s)$ is said to have a *positive decomposition* and it is externally positive: $\bar{G}(s) \bullet\!\!-\!\!\circ\, \bar{g}(t) \ge 0$ [16].

In the following, the controlled vehicle (17) will be analysed in order to find a set of closed-loop eigenvalues $\bar{\lambda}_i, i = 1, 2, \ldots, n{+}2$ that can be installed by a pole placement procedure so that the closed-loop transfer function $\bar{G}(s)$ has a positive decomposition.

B. Closed-Loop Transmission Zero

Although the vehicle model has no transmission zeros by Assumption 2, the controlled vehicle (17) has a transmission zero whose location depends on the controller parameters as shown by the following lemma.

Lemma 1: The controlled vehicle (17) has a single closed-loop transmission zero $\bar{\mu}$ at

$$\bar{\mu} = \frac{k_z}{k_d}. \tag{26}$$

Proof: Let $|\cdot|$ denote the determinant of a matrix. The zeros of the controlled vehicle (17) are determined by the roots of the determinant of the Rosenbrock system matrix as

$$|\boldsymbol{P}(\bar{\mu})| = \begin{vmatrix} \bar{\mu}\boldsymbol{I} - \bar{\boldsymbol{A}} & -\boldsymbol{e} \\ \bar{\boldsymbol{c}}^{\mathrm{T}} & 0 \end{vmatrix} \tag{27}$$

$$= \left|\begin{array}{cccc:c} \bar{\mu}\boldsymbol{I} - \boldsymbol{A} + \boldsymbol{b}\boldsymbol{k}_x^{\mathrm{T}} & \boldsymbol{b}k_d & \boldsymbol{b}k_z & & \boldsymbol{0} \\ \boldsymbol{c}^{\mathrm{T}} & \bar{\mu} & 0 & & -1 \\ -\beta\boldsymbol{c}^{\mathrm{T}} & 1 & \bar{\mu} & & 0 \\ \hdashline \boldsymbol{c}^{\mathrm{T}} & 0 & 0 & & 0 \end{array}\right| \tag{28}$$

$$= -1 \begin{vmatrix} \bar{\mu}\boldsymbol{I} - \boldsymbol{A} & 0 & \boldsymbol{b} \\ -\beta\boldsymbol{c}^{\mathrm{T}} & 1 & 0 \\ \boldsymbol{c}^{\mathrm{T}} & 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} \boldsymbol{I} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0}^{\mathrm{T}} & 1 & \bar{\mu} \\ -\boldsymbol{k}_x^{\mathrm{T}} & -k_d & -k_z \end{vmatrix} \tag{29}$$

$$= -1 \underbrace{\begin{vmatrix} \bar{\mu}\boldsymbol{I} - \boldsymbol{A} & -\boldsymbol{b} \\ \boldsymbol{c}^{\mathrm{T}} & 0 \end{vmatrix}}_{\neq 0,\ \forall \bar{\mu}} \cdot \begin{vmatrix} 1 & \bar{\mu} \\ -k_d & -k_z \end{vmatrix} = 0. \tag{30}$$

The first factor is non-zero for any $\bar{\mu}$ due to Assumptions 1–2 and the second factor is singular for the zero (26). □

The closed-loop transmission zero is important because of the following necessary conditions for external positivity.

Lemma 2 [15], [26]: Let λ_1 denote the dominant eigenvalue of a linear system, i.e.

$$\mathrm{Re}\{\lambda_i\} \le \mathrm{Re}\{\lambda_1\}, \quad i = 2, 3, \ldots, n. \tag{31}$$

If the considered system is externally positive, the dominant eigenvalue is real

$$\mathrm{Im}\{\lambda_1\} = 0 \tag{32}$$

and, furthermore, all real zeros μ_i are located to the left of the dominant eigenvalue λ_1 in the complex plane

$$\mu_i < \lambda_1, \quad \forall i. \tag{33}$$

Based on this result, the choice of the feedback gain $\boldsymbol{k}$ has to ensure that the transmission zero $\bar{\mu}$ is located sufficiently far in the left half plane to satisfy this necessary condition for external positivity. This requirement leads to a fundamental problem: It is well known in multivariable control theory that the closed-loop eigenvalues of a completely controllable system can be placed arbitrarily in the complex plane by a feedback of the state through one single control input. However, in order to place $\bar{\mu}$ arbitrarily, additional degrees of freedom are necessary in general. To circumvent this limitation, the strategy of the design procedure that will be proposed in the following is to place the zero and one closed-loop eigenvalue in the same location which renders Problem 1 solvable with the single control input of (6).

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C. Determination of the Feedback Gain

By using the pole placement approach, the feedback gain $\boldsymbol{k}$ is determined by Ackermann's formula (cf. [27])

$$\boldsymbol{k}^{\mathrm{T}} = \left(\boldsymbol{k}_{\mathrm{x}}^{\mathrm{T}}\ k_{\mathrm{d}}\ k_{\mathrm{z}}\right) = \boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}}\,\bar{P}(\boldsymbol{A}_0) \tag{34}$$

with the characteristic polynomial of the controlled vehicle

$$\bar{P}(\boldsymbol{A}_0) = \bar{a}_0\boldsymbol{I} + \bar{a}_1\boldsymbol{A}_0 + \cdots + \bar{a}_{n+1}\boldsymbol{A}_0^{n+1} + \boldsymbol{A}_0^{n+2} \tag{35}$$

and the vector

$$\boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}} = \left(0\ \cdots\ 0\ 1\right)\boldsymbol{S}_0^{-1}, \tag{36}$$

where the matrix

$$\boldsymbol{S}_0 = \left(\boldsymbol{b}_0\ \boldsymbol{A}_0\boldsymbol{b}_0\ \ldots\ \boldsymbol{A}_0^{n+1}\boldsymbol{b}_0\right) \tag{37}$$

denotes the controllability matrix of the open-loop model (15).

Instead of seeking the feedback gain $\boldsymbol{k}$ as a whole, an explicit representation of the parameters k_{d} and k_{z} that determine the closed-loop zero $\bar{\mu}$ is derived by decomposing Ackermann's formula as follows:

Lemma 3: The parameters k_{d} and k_{z} that determine the closed-loop transmission zero (26) are given by

$$k_{\mathrm{d}} = \frac{\bar{a}_0\beta - \bar{a}_1}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}} \quad \text{and} \quad k_{\mathrm{z}} = \frac{\bar{a}_0}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}}. \tag{38}$$

Proof: Given in Appendix A. □

D. Relation Between the Zero and the Eigenvalues

The goal of this section is to obtain a relation between the closed-loop zero $\bar{\mu}$ and the closed-loop eigenvalues in order to find a set of closed-loop eigenvalues that renders the controlled vehicle externally positive. With the result (38), the closed-loop zero (26) is determined by

$$\bar{\mu} = \frac{k_{\mathrm{z}}}{k_{\mathrm{d}}} = \frac{1}{\beta - \bar{a}_1/\bar{a}_0}. \tag{39}$$

In order to establish a connection between $\bar{\mu}$ and the closed-loop eigenvalues $\bar{\lambda}_i, (i = 1, 2, \ldots, n{+}2)$, consider the factorised form of the closed-loop characteristic polynomial

$$\bar{P}(\lambda) = \sum_{i=0}^{n+2} \bar{a}_i\,\lambda^i = \prod_{i=1}^{n+2}(\lambda - \bar{\lambda}_i) \tag{40}$$

with $\bar{a}_{n+2} = 1$. For a given set of eigenvalues $\bar{\lambda}_i$, the coefficients $\bar{a}_i$ of the polynomial are determined by Vieta's formulas [28]. The first two coefficients are given by

$$\bar{a}_0 = (-1)^{n+2}\prod_{i=1}^{n+2}\bar{\lambda}_i \tag{41}$$

$$\bar{a}_1 = (-1)^{n+1}\sum_{j=1}^{n+2}\prod_{\substack{i=1\\ i\neq j}}^{n+2}\bar{\lambda}_i, \tag{42}$$

which are sufficient to determine the feedback parameters (38) and the zero (39). Insertion of the coefficients (41) and (42) in (39) would result in a complicated expression. To avoid this, the following lemma provides a relation that significantly simplifies the fraction $\bar{a}_1/\bar{a}_0$ in (39).

Lemma 4: The following relation holds true:

$$\sum_{j=1}^{n+2}\prod_{\substack{i=1\\ i\neq j}}^{n+2}\bar{\lambda}_i = \prod_{i=1}^{n+2}\bar{\lambda}_i \cdot \sum_{j=1}^{n+2}\frac{1}{\bar{\lambda}_j}. \tag{43}$$

Proof: The products on the left-hand side can be extended by $\bar{\lambda}_j^{-1}$ which allows to drop the constraint $i \neq j$ in each product resulting in

$$\sum_{j=1}^{n+2}\prod_{\substack{i=1\\ i\neq j}}^{n+2}\bar{\lambda}_i = \sum_{j=1}^{n+2}\left(\prod_{i=1}^{n+2}\bar{\lambda}_i \cdot \frac{1}{\bar{\lambda}_j}\right). \tag{44}$$

Since the product of the right-hand side of (44) is independent of the index j, it can be rearranged to obtain the right-hand side of (43) which completes the proof. □

The following theorem summarises the previous results and provides a relation between the zero and the eigenvalues of the controlled vehicles $\bar{\Sigma}_i$.

Theorem 1: The eigenvalues $\bar{\lambda}_i, (i = 1, 2, \ldots, n{+}2)$ of the controlled vehicle (17) determine the zero $\bar{\mu}$ according to

$$\bar{\mu} = \frac{1}{\beta + \sum_{i=1}^{n+2}\bar{\lambda}_i^{-1}}. \tag{45}$$

Proof: Application of Lemma 4 to (41) and (42) yields

$$\frac{\bar{a}_1}{\bar{a}_0} = -\sum_{i=1}^{n+2}\frac{1}{\bar{\lambda}_i}, \tag{46}$$

which results in (45) after insertion in (39). □

E. Conditions for Closed-Loop External Positivity

This section presents conditions on the choice of the closed-loop eigenvalues $\bar{\lambda}_i$ so that the controlled vehicle (17) is asymptotically stable and externally positive for a given β. The fundamental requirement on the eigenvalues $\bar{\lambda}_i$ that ensures asymptotic stability is stated as

$$\mathrm{Re}\{\bar{\lambda}_i\} < \mathrm{Re}\{\bar{\lambda}_1\} < 0, \quad i = 2, 3, \ldots, n{+}2 \tag{47}$$

with $\bar{\lambda}_1$ denoting the dominant eigenvalue of the controlled vehicle. Theorem 1 shows that the choice of the time headway β and the closed-loop eigenvalues $\bar{\lambda}_i$ determine the transmission zero $\bar{\mu}$. Due to Lemma 2, this choice has to make sure that $\bar{\mu}$ is located to the left of the dominant eigenvalue. This problem will be solved by cancelling the closed-loop transmission zero with the smallest closed-loop eigenvalue:

$$\bar{\mu} \overset{!}{=} \bar{\lambda}_{n+2}. \tag{48}$$

It will be shown that this approach allows the controlled vehicle $\bar{\Sigma}_i$ to have a positive decomposition that renders it externally positive with appropriately chosen $\bar{\lambda}_i, (i = 1, 2, \ldots, n{+}2)$.

With (48), the relation (45) is equivalently stated as

$$\bar{\lambda}_{n+2}\left(\beta + \sum_{i=1}^{n+1}\frac{1}{\bar{\lambda}_i}\right) = 0 \tag{49}$$

and since $\bar{\lambda}_{n+2} < 0$ due to requirement (47), it follows that

$$\beta + \sum_{i=1}^{n+1}\frac{1}{\bar{\lambda}_i} = 0. \tag{50}$$

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By satisfying (50) with an appropriate choice of the closed-loop eigenvalues, the requirement (48) is satisfied.

Theorem 2: The controlled vehicle (17) satisfies the design objectives **(D1)** and **(D2)** if its eigenvalues $\bar{\lambda}_i, (i = 1, 2, \ldots, n{+}2)$ are chosen so as to satisfy the requirements

$$\mathrm{Re}\{\bar{\lambda}_i\} < \mathrm{Re}\{\bar{\lambda}_1\} < 0, \quad i = 2, 3, \ldots, n{+}2 \tag{51}$$

$$\mathrm{Im}\{\bar{\lambda}_i\} = 0, \quad i = 1, 2, \ldots, n{+}2 \tag{52}$$

$$\beta + \sum_{i=1}^{n+1} \frac{1}{\bar{\lambda}_i} = 0. \tag{53}$$

Proof: Equation (51) implies that **(D1)** is satisfied. The transfer function $\bar{G}(s)$ of the controlled vehicle is constructed as

$$\bar{G}(s) = \kappa \frac{s - \bar{\mu}}{\prod_{i=1}^{n+2}(s - \bar{\lambda}_i)} = \kappa \frac{1}{\prod_{i=1}^{n+1}(s - \bar{\lambda}_i)} \tag{54}$$

due to (48) with κ denoting the scalar factor

$$\kappa = \prod_{i=1}^{n+1}(-\bar{\lambda}_i) \tag{55}$$

due to (23). Thus, $\bar{G}(s)$ admits the positive decomposition

$$\bar{G}(s) = \prod_{i=1}^{n+1} \bar{G}_i(s) = \prod_{i=1}^{n+1} \frac{-\bar{\lambda}_i}{s - \bar{\lambda}_i} \tag{56}$$

with the correspondences

$$\bar{G}_i(s) \bullet\!\!-\!\!\circ\, \bar{g}_i(t) = -\bar{\lambda}_i \mathrm{e}^{\bar{\lambda}_i t} \geq 0, \quad t \geq 0, \tag{57}$$

which are nonnegative due to (51) and (52) and, hence, II-D is satisfied. □

The choice of the location of the zero is arbitrary apart from the condition $\bar{\mu} < \bar{\lambda}_1$. This is due to the fact that the eigenvalue $\bar{\lambda}_{n+2}$ and the zero cancel each other and, hence, do not appear in the transient behaviour. However, the presented conditions guarantee that $\bar{\lambda}_{n+2}$ is negative which is important since a pole/zero cancellation is not robust with respect to model uncertainties. Furthermore, the closed-loop transfer function (24) can be reformulated to contain the element

$$\bar{G}_1(s) = \frac{s - \bar{\mu}}{s - \bar{\lambda}_1} \bullet\!\!-\!\!\circ\, \bar{g}_1(t) = (\bar{\lambda}_1 - \bar{\mu})\, \mathrm{e}^{\bar{\lambda}_1 t} \geq 0, \tag{58}$$

which is externally positive due to (47) and (48). Suppose that $\bar{\mu}$ is not equal to $\bar{\lambda}_{n+2}$ due to model uncertainties. The closed-loop system will still have a positive decomposition as long as $\bar{\mu} < \bar{\lambda}_1$ holds true. Thus, a small deviation

$$|\bar{\mu} - \bar{\lambda}_{n+2}| < \epsilon \tag{59}$$

with $\epsilon > 0$ does not jeopardise the stability or external positivity of the controlled vehicle.

Note that Theorem 2 can be relaxed to allow complex conjugate non-dominant eigenvalues, i.e. condition (52) can be replaced by $\mathrm{Im}\{\bar{\lambda}_1\} = 0$ since it is only necessary that the dominant eigenvalue is real.

F. Closed-Loop Eigenvalues That Achieve External Positivity

This section will present an explicit recursive procedure to find a set of closed-loop eigenvalues that satisfies the conditions of Theorem 2. The eigenvalues have to satisfy the requirement (53), which is written out as

$$\underbrace{\beta}_{>0} + \underbrace{\frac{1}{\bar{\lambda}_1} + \frac{1}{\bar{\lambda}_2} + \cdots + \frac{1}{\bar{\lambda}_{n+1}}}_{<0} = 0, \tag{60}$$

where $\bar{\lambda}_1$ should be the dominant eigenvalue (cf. (47)). An upper bound for the dominant eigenvalue is given by

$$\bar{\lambda}_1 < -\frac{1}{\beta}, \tag{61}$$

which guarantees that the subtotal $\beta + \bar{\lambda}_1^{-1}$ is positive. Otherwise, (60) has no solution since all eigenvalues have the same sign. Thus, for a fixed $\bar{\lambda}_1$, the requirement

$$\underbrace{\beta + \frac{1}{\bar{\lambda}_1}}_{>0} + \underbrace{\frac{1}{\bar{\lambda}_2} + \cdots + \frac{1}{\bar{\lambda}_{n+1}}}_{<0} = 0 \tag{62}$$

has to be satisfied by choosing $\bar{\lambda}_2$ according to

$$\bar{\lambda}_2 < -\frac{1}{\beta + \bar{\lambda}_1^{-1}}. \tag{63}$$

This procedure can be repeated for the remaining eigenvalues which leads to the recursion rule

$$\bar{\lambda}_i < -\frac{1}{\beta + \sum_{j=1}^{i-1} \bar{\lambda}_j^{-1}}, \quad i = 2, 3, \ldots, n. \tag{64}$$

The last eigenvalue is determined by

$$\bar{\lambda}_{n+1} = -\frac{1}{\beta + \sum_{j=1}^{n} \bar{\lambda}_j^{-1}} \tag{65}$$

to meet the condition (60) exactly.

Choosing the closed-loop eigenvalues according to this procedure will make sure that the requirement (53) is satisfied so that the objective (48) is achieved. In order to guarantee that $\bar{\lambda}_1$ is the dominant eigenvalue (to satisfy (51)), a lower bound has to be imposed: The upper bound (63) of $\bar{\lambda}_2$ has to be located to the left of $\bar{\lambda}_1$ in the complex plane so that there is no eigenvalue with a larger real part than $\bar{\lambda}_1$, i.e.

$$\bar{\lambda}_2 < -\frac{1}{\beta + \bar{\lambda}_1^{-1}} < \bar{\lambda}_1, \tag{66}$$

which is satisfied if and only if $-\frac{2}{\beta} < \bar{\lambda}_1$. Hence, the feasible region for $\bar{\lambda}_1$ to be the dominant eigenvalue is

$$-\frac{2}{\beta} < \bar{\lambda}_1 < -\frac{1}{\beta}. \tag{67}$$

The developed procedure to find closed-loop eigenvalues that render the controlled vehicle asymptotically stable and externally positive is summarised in Algorithm 1.

Note that the eigenvalues obtained by Algorithm 1 only represent one possible solution to satisfy the requirements of Theorem 2. Another possibility is to place all closed-loop

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Algorithm 1 Eigenvalues That Achieve External Positivity

Input: Desired time headway β and dynamical order n of Σ_i.

1: Choose the dominant eigenvalue in the given interval:

$$\bar{\lambda}_1 \in \left(-\frac{2}{\beta}, -\frac{1}{\beta}\right). \tag{68}$$

2: For $i = 2, 3, \ldots, n$, choose $\bar{\lambda}_i$ iteratively with

$$\bar{\lambda}_i < -\frac{1}{\beta + \sum_{j=1}^{i-1} \bar{\lambda}_j^{-1}}. \tag{69}$$

3: Determine the eigenvalue

$$\bar{\lambda}_{n+1} = -\frac{1}{\beta + \sum_{j=1}^{n} \bar{\lambda}_j^{-1}}. \tag{70}$$

4: Choose the last eigenvalue according to

$$\bar{\lambda}_{n+2} < \bar{\lambda}_1. \tag{71}$$

Result: Eigenvalues $\bar{\lambda}_i$, $(i = 1, 2, \ldots, n{+}2)$ with which the controlled vehicle (17) satisfies **(D1)** and **(D2)**.

eigenvalues in the same location. The condition (53) is then satisfied if

$$\bar{\lambda}_i = -\frac{n+1}{\beta}, \quad i = 1, 2, \ldots, n+1 \tag{72}$$

holds true. However, for a large system order n, the eigenvalues are shifted far to the left in the complex plane which restricts the range of feasible values for β if control input boundaries are considered. The eigenvalues determined by Algorithm 1 are more widely distributed on the real axis but the dominant eigenvalue $\bar{\lambda}_1$ is allowed to be slower, i. e. located closer to the origin of the complex plane which allows for implementing a smaller time headway β.

G. Summary of the Design Procedure

Given the closed-loop eigenvalues found by Algorithm 1, the coefficients $\bar{a}_i$ of the characteristic polynomial are determined by (40). The feedback gain $\boldsymbol{k}$ that solves Problem 1 is then determined by Ackermann's formula (34) as stated by the following theorem.

Theorem 3: If the eigenvalues $\bar{\lambda}_i$, $(i = 1, 2, \ldots, n{+}2)$ that are chosen according to Algorithm 1 are placed by the state feedback gain $\boldsymbol{k}$ via Ackermann's formula, the controlled vehicle (17) satisfies the design objectives **(D1)** and **(D2)**.

Proof: Choosing the dominant eigenvalue from the interval (68) ensures that $\bar{\lambda}_1$ is real and negative and the other eigenvalues are located to the left of it due to the recursion rule (69). Hence, (51) and (52) are satisfied. Furthermore, the choice of the remaining eigenvalues $\bar{\lambda}_i$, $(i = 2, 3, \ldots, n{+}2)$ by (69)–(71) guarantees that condition (53) is satisfied. Thus, the controlled vehicles $\bar{\Sigma}_i$ have a positive decomposition and are asymptotically stable which proves the theorem and correctness of Algorithm 1. □

The complete design procedure is summarised by Algorithm 2. Note that the vehicle model (6) does not have to be identical for each vehicle i. Thus, a platoon of vehicles with controllers that are designed individually by the following algorithm will satisfy the requirements **(R1)**–**(R4)**.

Algorithm 2 Design of Externally Positive Vehicles

Input: $\Sigma_i = (\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{c}^{\mathrm{T}})$ and desired time headway β.

1: Execute Algorithm 1 with $n = \dim \boldsymbol{A}$ and β.

2: Calculate the coefficients $\bar{a}_i$, $(i{=}0, 1, \ldots, n{+}1)$ with (40).

3: Construct $\boldsymbol{A}_0$ and $\boldsymbol{S}_0$ according to (15) and (37).

4: Determine $\boldsymbol{k}$ with Ackermann's formula (34).

Result: Feedback gain $\boldsymbol{k}$ that renders $\bar{\Sigma}_i$ asymptotically stable and externally positive.

IV. Application Example

A. Standard Vehicle Model

Consider a vehicle represented by

$$G(s) = \frac{1}{s(\tau s + 1)} \tag{73}$$

with the minimal state-space realisation

$$\Sigma_i : \begin{cases} \begin{pmatrix} \dot{v}_i(t) \\ \dot{a}_i(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -\frac{1}{\tau} \end{pmatrix} \begin{pmatrix} v_i(t) \\ a_i(t) \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{\tau} \end{pmatrix} u_i(t) \\ v_i(t) = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} v_i(t) \\ a_i(t) \end{pmatrix} \end{cases} \tag{74}$$

with $a_i(t)$ denoting the acceleration of the vehicle. This model is widely used in the literature of vehicle platooning (cf. [9]–[13] for example) and satisfies Assumption 2. The time constant for the test vehicles in [10] was identified as $\tau = 100\,\mathrm{ms}$. The open-loop dynamics (15) read as

$$\begin{pmatrix} \dot{v}_i(t) \\ \dot{a}_i(t) \\ \dot{d}_i(t) \\ \dot{z}_i(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{1}{\tau} & 0 & 0 \\ -1 & 0 & 0 & 0 \\ \beta & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} v_i(t) \\ a_i(t) \\ d_i(t) \\ z_i(t) \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{\tau} \\ 0 \\ 0 \end{pmatrix} u_i(t) + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} v_{i-1}(t). \tag{75}$$

The procedure presented in this paper will be applied to find a feedback gain $\boldsymbol{k}$ that renders the controlled vehicle asymptotically stable and externally positive via

$$u_i(t) = -\boldsymbol{k}^{\mathrm{T}} \begin{pmatrix} v_i(t) \\ a_i(t) \\ d_i(t) \\ z_i(t) \end{pmatrix} \tag{76}$$

for a given time headway coefficient, e. g. $\beta = 1\,\mathrm{s}$. Note that the state consisting of the velocity $v_i(t)$, the acceleration $a_i(t)$, the inter-vehicle distance $d_i(t)$ as well as the regulator state $z_i(t)$ are quantities that are measurable in modern vehicles and, hence, there is no need for an observer or estimator.

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B. Controller Design

Ackermann's formula (34) leads to

$$\boldsymbol{k}^{\mathrm{T}} = \begin{pmatrix} \bar{a}_0 & \bar{a}_1 & \bar{a}_2 & \bar{a}_3 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & \beta\tau & \tau \\ 0 & 0 & -\tau & 0 \\ \tau & 0 & 0 & 0 \\ 0 & \tau & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \tag{77}$$

In order to determine the coefficients $\bar{a}_i$, Algorithm 1 is executed with $n = 2$ and $\beta = 1\,\mathrm{s}$ as follows: At first, the dominant closed-loop eigenvalue $\bar{\lambda}_1$ is chosen from the interval $(-2, -1)$ as follows:

$$\bar{\lambda}_1 = -1.5\,\mathrm{s}^{-1}. \tag{78}$$

This choice results in the boundary $\bar{\lambda}_2 < -3\,\mathrm{s}^{-1}$ which is satisfied by

$$\bar{\lambda}_2 = -5\,\mathrm{s}^{-1}. \tag{79}$$

The second to last eigenvalue is fixed by (70) as

$$\bar{\lambda}_3 = -7.5\,\mathrm{s}^{-1} \tag{80}$$

and the last eigenvalue which is equal to the transmission zero is, apart from the requirement $\bar{\mu} < \bar{\lambda}_1$, free to choose, e. g. as

$$\bar{\lambda}_4 = \bar{\mu} = -9\,\mathrm{s}^{-1}. \tag{81}$$

Using (40), the coefficients $\bar{a}_i$ are calculated which fully determines the right-hand side of (77) and results in

$$\boldsymbol{k}^{\mathrm{T}} = \begin{pmatrix} 18.225 & 1.300 & -5.625 & 50.625 \end{pmatrix}. \tag{82}$$

The impulse response $\bar{g}(t) \circ\!\!-\!\!\bullet \bar{G}(s)$ is shown in Fig. 4 to verify that the controlled vehicle is indeed externally positive.

V. Experimental Study

A. Experimental Environment

The proposed design method has been tested in the experimental environment illustrated in Fig. 3. A set of N identical differentially driven robots will be considered that should travel in a platoon along a circular path. To this end, a low-level path controller is applied that guides the robots along the prescribed path (for a detailed description cf. [29]). The behaviour of the robots along the path is governed by a linear model and, hence, Algorithm 2 can be used to design the platooning controller as described in the last section.

B. Experimental Results

This section will present two measurements: First, the impulse response of a controlled robot is measured to show that the robot is indeed rendered externally positive by Algorithm 2 and, second, a platoon of nine robots with a changing reference velocity of the leader is examined. Since the impulse response is hardly measurable directly, it will be obtained with a trick. The unforced (free) motion of a controlled robot (17) is determined by

$$v_{i,\mathrm{free}}(t) = \bar{\boldsymbol{c}}^{\mathrm{T}} \mathrm{e}^{\bar{A}t} \bar{\boldsymbol{x}}_{i0} \tag{83}$$

with $\bar{\boldsymbol{x}}_{i0}^{\mathrm{T}} = \begin{pmatrix} \boldsymbol{x}_{i0}^{\mathrm{T}} & d_{i0} & z_{i0} \end{pmatrix}$, which differs only in the last factor from its impulse response

$$\bar{g}(t) = \bar{\boldsymbol{c}}^{\mathrm{T}} \mathrm{e}^{\bar{A}t} \boldsymbol{e}. \tag{84}$$

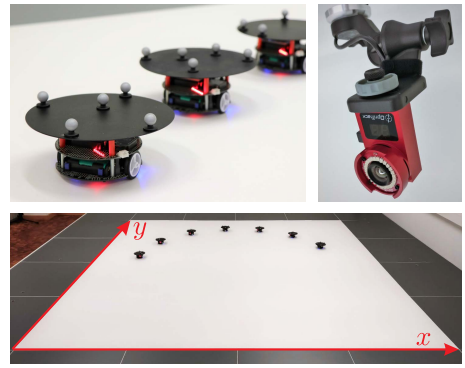

Fig. 3. Robots, cameras, and driving surface of the experimental plant.

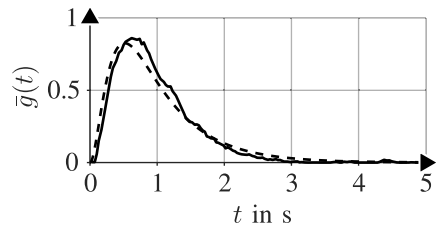

Fig. 4. Simulated (dashed) and measured (solid) impulse response.

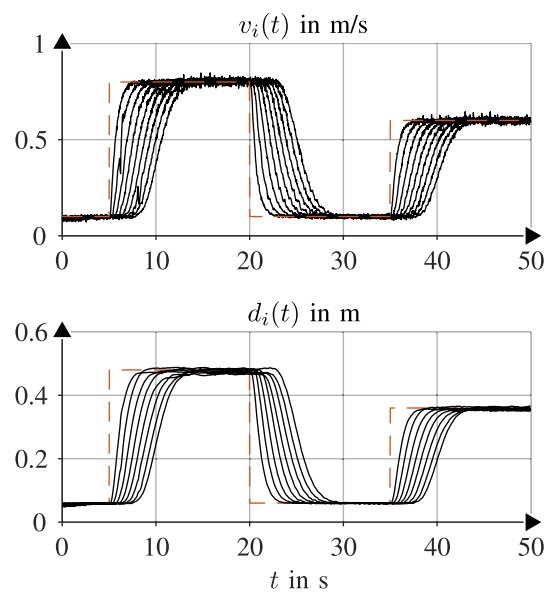

Fig. 5. Experimental evaluation of a platoon of $N = 9$ robots.

Thus, the impulse response $\bar{g}(t)$ of the controlled robot can be obtained by measuring its velocity $v_i(t)$ with $v_{i-1}(t) \equiv 0$ if the initial state is set according to

$$\begin{pmatrix} \boldsymbol{x}_{i0} \\ d_{i0} \\ z_{i0} \end{pmatrix} = \begin{pmatrix} \boldsymbol{0} \\ 1\,\mathrm{m} \\ 0 \end{pmatrix} = \boldsymbol{e} \tag{85}$$

with $\boldsymbol{e}$ defined in (15). The result is depicted in Fig. 4 which shows that the closed-loop impulse response is nonnegative for all t, i. e. the proposed design procedure satisfies the design goal in an experimental environment.

Figure 5 shows the measurement of a set of $N = 9$ identical robots each equipped with feedback controllers that are designed as demonstrated in the last section. In the top graph, the velocity $v_i(t)$ is shown. The piecewise constant reference velocity v_0 (depicted by the dashed line) changes three times

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to different levels. It can be seen that after each transition, all robots travel with the correct velocity. Furthermore, the transition is monotonic due to the externally positive dynamics. The bottom graph shows the inter-robot distance $d_i(t)$. The reference time headway βv_0 is depicted by the dashed line. Clearly, the inter-robot distances approach the reference value in a steady state without any overshoot similarly to the velocities. Consequently, there are no collisions in the platoon since $d_i(t) > 0$ holds true at any time. The measurements show that the requirements **(R1)**–**(R4)** are satisfied by the platoon with the controlled robots (17) and the feedback gain $\boldsymbol{k}$ designed by the proposed method.

VI. Conclusion

The present paper addressed the design of adaptive cruise controllers that satisfy specific design objectives for a general class of linear vehicle models without transmission zeros. The main result is a procedure that solves the design problem and, particularly, renders the controlled vehicles externally positive using an extended state feedback. For that purpose, an algorithm was developed that finds a set of closed-loop eigenvalues that corresponds to an externally positive system which are then implemented using pole placement techniques. The results are applied to a standard model in the literature on ACC and an experimental evaluation with mobile robots is given. The procedure has been shown to satisfy the design goals **(D1)** and **(D2)** and, hence, to achieve the desired platoon behaviour **(R1)**–**(R4)** in the experimental environment.

Appendix A
Proof of Lemma 3

The lemma is proved in two steps. Both the vector $\boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}}$ and the characteristic polynomial $\bar{P}(\boldsymbol{A}_0)$ are decomposed in the following to obtain the result (38).

Decomposition of the Controllability Matrix: Assumption 2 will be applied to decompose $\boldsymbol{S}_0$ (cf. (86), as shown at the bottom of the page) as follows. The relative degree r of a system without zeros is equal to its order: $r = n$. As a consequence, the Markov parameters of the vehicle model Σ_i satisfy the properties (cf. [30])

$$\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{i}\boldsymbol{b} = 0, \quad i = 0, 1, \ldots, n-2 \tag{87}$$

$$\text{and} \quad \boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b} \neq 0. \tag{88}$$

Application of (87) and (88) to $\boldsymbol{S}_0$ yields a decomposition in block triangular form as

$$\boldsymbol{S}_0 = \begin{pmatrix} \boldsymbol{S} & \boldsymbol{R} \\ \boldsymbol{O} & \boldsymbol{Q} \end{pmatrix} \tag{89}$$

with the controllability matrix $\boldsymbol{S}$ of the vehicle model (6) defined in (9) and

$$\boldsymbol{R} = \begin{pmatrix} \boldsymbol{A}^{n}\boldsymbol{b} & \boldsymbol{A}^{n+1}\boldsymbol{b} \end{pmatrix} \in \mathbb{R}^{n\times 2} \tag{90}$$

$$\boldsymbol{Q} = \begin{pmatrix} -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b} & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n}\boldsymbol{b} \\ \beta\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b} & \boldsymbol{c}^{\mathrm{T}}(\boldsymbol{I}+\beta\boldsymbol{A})\boldsymbol{A}^{n-1}\boldsymbol{b} \end{pmatrix} \in \mathbb{R}^{2\times 2}. \tag{91}$$

The $2\times n$ zero matrix $\boldsymbol{O}$ in the bottom left of (89) results from the fact that the first n elements in the last two rows of (86) are equal to zero due to (87).

With the decomposition of $\boldsymbol{S}_0$, the determination of the last row of its inverse in (36) is equivalently stated as

$$\boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}}\boldsymbol{S}_0 = \begin{pmatrix} 0 & \cdots & 0 & 1 \end{pmatrix} \tag{92}$$

$$\iff \begin{pmatrix} \boldsymbol{s}_{\mathrm{R}}^{\mathrm{T}} & \boldsymbol{q}^{\mathrm{T}} \end{pmatrix} \begin{pmatrix} \boldsymbol{S} & \boldsymbol{R} \\ \boldsymbol{O} & \boldsymbol{Q} \end{pmatrix} = \begin{pmatrix} 0 & \cdots & 0 & 1 \end{pmatrix} \tag{93}$$

by introduction of

$$\boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}} = \begin{pmatrix} \boldsymbol{s}_{\mathrm{R}}^{\mathrm{T}} & \boldsymbol{q}^{\mathrm{T}} \end{pmatrix}. \tag{94}$$

Equation (93) is then decomposed into

$$\boldsymbol{s}_{\mathrm{R}}^{\mathrm{T}}\boldsymbol{S} = \boldsymbol{0}^{\mathrm{T}} \tag{95}$$

$$\boldsymbol{s}_{\mathrm{R}}^{\mathrm{T}}\boldsymbol{R} + \boldsymbol{q}^{\mathrm{T}}\boldsymbol{Q} = \begin{pmatrix} 0 & 1 \end{pmatrix}. \tag{96}$$

The first equation holds true if and only if

$$\boldsymbol{s}_{\mathrm{R}}^{\mathrm{T}} = \boldsymbol{0}^{\mathrm{T}} \tag{97}$$

since $\boldsymbol{S}$ is regular due to Assumption 1. Hence, the second equation simplifies to

$$\boldsymbol{q}^{\mathrm{T}}\boldsymbol{Q} = \begin{pmatrix} 0 & 1 \end{pmatrix}, \tag{98}$$

which can be solved by straightforward inversion of $\boldsymbol{Q}$:

$$\boldsymbol{q}^{\mathrm{T}} = \begin{pmatrix} 0 & 1 \end{pmatrix}\boldsymbol{Q}^{-1} = \frac{1}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}}\begin{pmatrix} \beta & 1 \end{pmatrix}. \tag{99}$$

In summary, the first factor of Ackermann's formula (34) is

$$\boldsymbol{s}_{\mathrm{R0}}^{\mathrm{T}} = \left(0 \ldots 0 \ \frac{\beta}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}} \ \frac{1}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}}\right). \tag{100}$$

Determination of the Feedback Gain: In order to evaluate the characteristic polynomial (35) of the controlled vehicle, the m-th power of $\boldsymbol{A}_0$ in (15)

$$\boldsymbol{A}_0^{m} = \left(\begin{array}{c:c} \boldsymbol{A}^{m} & \boldsymbol{0} \ \boldsymbol{0} \\ \hdashline -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{m-1} & 0 \ 0 \\ \boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{m-2}(\beta\boldsymbol{A}+\boldsymbol{I}) & 0 \ 0 \end{array}\right), \quad m > 1 \tag{101}$$

has to be examined. Equation (101) shows that the last two columns are equal to zero for exponents $m > 1$. Thus, the characteristic polynomial (35) has the block triangular form

$$\bar{P}(\boldsymbol{A}_0) = \begin{pmatrix} \bar{P}(\boldsymbol{A}) & \boldsymbol{O} \\ * & \boldsymbol{T} \end{pmatrix} \tag{102}$$

$$\boldsymbol{S}_0 = \left(\begin{array}{cccc:cc} \boldsymbol{b} & \boldsymbol{A}\boldsymbol{b} & \boldsymbol{A}^2\boldsymbol{b} & \cdots & \boldsymbol{A}^{n-1}\boldsymbol{b} & \boldsymbol{A}^{n}\boldsymbol{b} & \boldsymbol{A}^{n+1}\boldsymbol{b} \\ \hdashline 0 & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{b} & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{b} & \cdots & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-2}\boldsymbol{b} & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b} & -\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n}\boldsymbol{b} \\ 0 & \beta\boldsymbol{c}^{\mathrm{T}}\boldsymbol{b} & \boldsymbol{c}^{\mathrm{T}}(\boldsymbol{I}+\beta\boldsymbol{A})\boldsymbol{b} & \cdots & \boldsymbol{c}^{\mathrm{T}}(\boldsymbol{I}+\beta\boldsymbol{A})\boldsymbol{A}^{n-3}\boldsymbol{b} & \boldsymbol{c}^{\mathrm{T}}(\boldsymbol{I}+\beta\boldsymbol{A})\boldsymbol{A}^{n-2}\boldsymbol{b} & \boldsymbol{c}^{\mathrm{T}}(\boldsymbol{I}+\beta\boldsymbol{A})\boldsymbol{A}^{n-1}\boldsymbol{b} \end{array}\right) \tag{86}$$

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with $*$ denoting a non-zero $2\times n$ matrix and

$$\boldsymbol{T} = \begin{pmatrix} \bar{a}_0 & 0 \\ -\bar{a}_1 & \bar{a}_0 \end{pmatrix}, \tag{103}$$

which results from the bottom right 2×2 matrix of the first two summands of $\bar{P}(\boldsymbol{A}_0)$. Applying (94), (97) and (102) to Ackermann's formula (34) yields

$$\begin{pmatrix} \boldsymbol{k}_{\mathrm{x}}^{\mathrm{T}} & k_{\mathrm{d}} & k_{\mathrm{z}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{0}^{\mathrm{T}} & \boldsymbol{q}^{\mathrm{T}} \end{pmatrix} \begin{pmatrix} \bar{P}(\boldsymbol{A}) & \boldsymbol{O} \\ * & \boldsymbol{T} \end{pmatrix}. \tag{104}$$

Consequently, the parameters k_{d} and k_{z} that determine the closed-loop transmission zero (26) are given by

$$\begin{pmatrix} k_{\mathrm{d}} & k_{\mathrm{z}} \end{pmatrix} = \boldsymbol{q}^{\mathrm{T}}\boldsymbol{T} \tag{105}$$

or, explicitly, by

$$k_{\mathrm{d}} = \frac{\bar{a}_0\beta - \bar{a}_1}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}} \quad \text{and} \quad k_{\mathrm{z}} = \frac{\bar{a}_0}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{A}^{n-1}\boldsymbol{b}} \tag{106}$$

using the result (99), which completes the proof.

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